Rajveer Gupta

+91-8563938645 • rajveercareer1@gmail.com • LinkedIn • GitHub • Kaggle • Hugging Face

Portfolio: live demos, project case studies and the IEEE paper

Education

Indian Institute of Information Technology, Nagpur Aug 2024 – Jun 2028 (expected)
B.Tech, Computer Science & Engineering — CGPA: 9.08/10 *Nagpur, India*
Coursework: DSA, OOP (AA), Software Engineering (AA), Operating Systems (AB), Programming Language Design (AB), DBMS

Publication

A. Gupta, A. Singh, **R. Gupta**, P. Jain, T. Patidar, "Fault Tolerant Industrial Vision Framework Through Semantic-Guided Cross-View Temporal Reconstruction and Restoration," *IEEE IRAI 2026*, Melbourne (accepted and presented) | **Paper**

Experience

Research Intern, Computer Vision — IIIT Nagpur (Advisor: Dr. Nishat Afshan Ansari) 2026
HocVid · PyTorch, OpenCV, NoPoSplat (3D Gaussian Splatting), CoTracker2, SAM, DiffuEraser, MiDaS, Mamba
Overview: co-developed HocVid, which detects a dead camera in a **3**-camera network and rebuilds the lost view from overlapping cameras and the camera's last **16** frames.

- Detected camera faults with a degradation score (frequency anomalies, cross-camera SSIM, hardware flags) that labels cameras healthy, degraded or dead; a 16-frame **hysteresis** window stops flicker: **89.4% accuracy**.
- Calibrated cameras with SIFT + RANSAC (fundamental matrix, intrinsic and extrinsic parameters); rebuilt surviving views as **pose-free 3D Gaussians** (NoPoSplat) from the failed camera's stored pose, worth **1.85 dB**.
- Evaluated under injected physical faults (blur, lens dust, thermal noise, signal loss): **30.02 dB PSNR** on Wildtrack, +3.65 dB over the best naive cascade; **32.42 dB / 0.953 SSIM** on GoPro deblurring, **+2.37 dB** over MoCE-IR.

Projects

ReturnGuard — Multi-Agent Return-Fraud Detection for E-commerce 2026
GitHub | Live | Demo — Python, LangGraph, Gemini, PyTorch, DINOv2, Pydantic, Flask, SQLite, GitHub Actions
Overview: checks at the doorstep that a returned product is the unit that was delivered, before the refund.

- Designed a LangGraph state machine with parallel policy, history and vision agents and an LLM-free triage router: **79%** of returns need zero LLM calls (**0.21** per return, about **$1** per **1,000**).
- Fixed the vision signal: **65–77%** of DINOv2 patches were square padding, so object-only pooling cut a false "similar" score from **0.77** to **0.38**; a colour check catches same-model colour swaps.
- Measured with confidence intervals: **100%** precision, **93.3%** recall and **0.986** F0.5 on **48** labeled pairs (v1: **90.9%** precision); **97.4%** precision at **100%** recall across **36** photo-distortion conditions.
- Added a blind label-reading agent, Pydantic-validated outputs, timeouts and retries with a fail-safe to human review; **35** offline tests run in CI, which auto-deploys to Hugging Face Spaces.

Memora — Memory-Care App for Alzheimer's Patients 2025 – 2026
GitHub | Live | Results | Case study — Python, Flask, ONNX, PyTorch, RoBERTa · Sole developer
Overview: tells a patient who is in front of them in a loved one's voice and finds medicines; rebuilt solo from ARKA.

- Raised LFW face recognition from **87.1%** to **98.9%** (95% CI **98.2–99.5**) and cut latency from **395** to **21 ms** by fixing detection and alignment first (YuNet + SFace).
- Lifted distress-detection F1 from **0.39** to **0.88** with RoBERTa (GoEmotions) and trained a MiniLM intent classifier at **82.6%** accuracy (keyword rules: **36%**), published on the Hugging Face Hub.
- Secured the app with AES-256-GCM encryption at rest, CSRF and XSS regression tests and a consent-gated AWS Rekognition fallback; **81** offline tests in CI and **53/53** end-to-end checks on the live deployment.

Resolv — Multilingual Entity Resolution over 11.7M Records (Amazon ML Challenge 2026) 2026
*GitHub | Live | Demo — Team lead of 4 · **AIR 471 of 32K teams** · PyTorch, multilingual-e5, Qwen3, LightGBM*

- Combined IDF-weighted keys from **11** families with a multilingual-e5 bi-encoder fine-tuned with InfoNCE: **99.86%** pair recall; a LightGBM re-ranker cut **86.7** candidates per business to **5.58**.
- Stacked **107** hand-built features with a cross-encoder and a fine-tuned **Qwen3-Reranker-0.6B** on uncertain pairs, lifting leaderboard F0.5 from **0.94** to **0.986** (**0.998** validation precision).

Achievements

- **Led the team** in **Amazon ML Challenge 2026** to **AIR 471 of 32K teams**.
- **Amazon ML Summer School 2026**, selected in the **top 3,000 of 134,000+** applicants.
- **Led the team** in **Microsoft Imagine Cup 2025** to the **MVP Round** (**60K+** teams); managed the full ML pipeline.
- **Smart India Hackathon 2025**, **National Semi-Finalist** (AyurSutra) among **100,000+** participants.
- **Best Innovation Award**, IIT Hyderabad (Memora) • IEEE IES GenAI Hackathon Finalist.

Skills

LLMs & Agents: LangGraph, LangChain, Gemini (structured outputs), Pydantic, RAG, Hugging Face
ML & Deep Learning: PyTorch, Transformers, RoBERTa, Whisper, DINOv2, ONNX Runtime, scikit-learn, LightGBM
Programming & Tools: Python, C++, C, Java, SQL; Flask, FastAPI, Docker, Git, Linux
MLOps: GitHub Actions CI/CD, MLflow, Hugging Face Spaces, AWS (Rekognition, Location Service)